



Cobalt containing crystallizing glass seals for solid oxide fuel cells – A new strategy for strong adherence to metals and high thermal expansion



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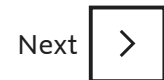
Highlights

- Cobalt and barium containing silicate glasses were prepared.
- The powdered glasses could be fully densified during sintering.
- At the end of sintering barium and cobalt containing crystals were precipitated.

- At the interface, cobalt ions were reduced by the nicrofer alloy to metallic cobalt.
- The glasses exhibit high thermal expansion and strong adherence to the metal.

Abstract

Powdered glasses based on the systems BaO/CoO/SiO₂ and BaO/ZnO/CoO/SiO₂ with small additions of ZrO₂, La₂O₃ and B₂O₃ were investigated with respect to their sintering and crystallization behavior. They could be fully densified by viscous flow and during subsequent crystallization BaSi₂O₅ and Ba(Zn_xCo_{2-x})Si₂O₇ or BaCo₂Si₂O₇ were formed. The CTEs (100–800°C) of crystallized samples were $14.0\text{--}14.6\times 10^{-6}\text{K}^{-1}$. Sealing experiments with Nicrofer[®] alloy showed strong adherence of the crystallized glass. Aluminum from the alloy is oxidized and forms an alumina layer or interconnected structures inside the metal. Cobalt oxide as a component of the glass is reduced to the metal and occurs as approximately spherical particles near the Nicrofer[®] alloy. The number of cobalt particles increases with increasing sealing temperature. During fracture of sealed Nicrofer[®] plates, the crack runs through the glass and not through the interface metal/crystallized glass.



Keywords

Solid oxide fuel cells; Crystallizing glass seals; Thermal expansion; Barium silicates; Barium cobalt zinc silicates

1. Introduction

Joining of materials, such as metals or ceramics [1] is frequently done by glasses. This is of great importance especially for solid oxide fuel cells (SOFCs) and other chemical reactors which are used at high temperatures typical in the range from 700 to 1000°C [2]. For this purpose, the glass is powdered and brought in between the materials to be joined. Then this arrangement is thermally annealed which leads to a densification (sintering) by viscous flow.

Unfortunately, there seems to be a certain correlation between the coefficient of thermal expansion (CTE) and the glass transition temperature, T_g (or the softening temperature) [1]. Usually, glasses with high T_g ($>750^\circ\text{C}$) possess fairly low CTE [3], [4], while glasses with high CTE ($>12 \times 10^{-6} \text{K}^{-1}$) exhibit low T_g and hence fairly small softening temperatures [5], [6], [7], [8], [9].

In the case of solid oxide fuel cells or other high temperature reactors, metals are to be joint, which commonly have CTEs $>11 \times 10^{-6} \text{K}^{-1}$ [10], [11] and the operating temperatures are in the range from 700 to 1000°C [12]. Glasses, however, which fulfill these two requirements, are not available and seals which during applications contain only amorphous phases cannot be applied.

Seals which might fulfill these two requirements do not only contain a glassy phase, but additionally at least one crystalline phase. Such materials, in principle, might be prepared using two different ways: a crystalline phase of high thermal expansion coefficient is given to the glass and then thermal treatment is carried out which leads to densification by viscous flow [13]. This procedure works well if the volume percentage of crystalline phase is not too high ($<30\%$) because otherwise densification is difficult to achieve [14]. The second possible way is to precipitate a crystalline phase from the glass after the densification by sintering is completed [15]. The starting material in this case is a homogeneous glass. Materials prepared according these two different ways are usually denominated as “composite seals” or “crystallizing seals”, respectively [2], [16].

For crystallizing seals, a glass is melted, crushed and powdered. Then it is brought between the materials to be joined. During subsequent thermal treatment, crystallization should not occur in a notable extend until complete densification is achieved [17]. Subsequently, the seal is crystallized. In order to accelerate crystallization, the latter step might also be carried out at another temperature than the densification step.

In summary, the glass composition and the temperature/time schedule have to be optimized in order to control densification and crystallization. Furthermore, the grain size and grain size distribution have to be adjusted to the respective glass composition, procedure and application.

Crystallizing glasses for solid-oxide fuel cells must be stable at high temperatures (typically $\geq 850^\circ\text{C}$) over a long period of time (many years). The softening temperature of the seal should be above the operating temperature [18], [19]. The CTEs of the crystallized seal and of the different components of the cell should be in agreement within $\pm 1 \cdot 10^{-6}\text{K}^{-1}$ [18].

In SOFCs as cathode mostly perovskites such as $\text{La}_{1-x}\text{Sr}_x\text{CoO}_{3\pm\delta}$ are used [20]. Their CTEs ($30\text{--}800^\circ\text{C}$) are typical in the range from 11 to $14 \times 10^{-6}\text{K}^{-1}$ [19], [21]. As anode, a cermet of nickel and stabilized zirconia is used. The CTE ($25\text{--}1000^\circ\text{C}$) is in the range from 12.7 to $13.3 \times 10^{-6}\text{K}^{-1}$ for a nickel concentration of $30\text{--}45\text{wt}\%$ [22]. The interconnect in SOFCs is commonly composed by different heat- and oxidation resistant materials such as Ni-based alloys [23], exhibiting CTEs in the range from 14 to $19 \times 10^{-6}\text{K}^{-1}$ [10].

Besides a good matching of CTEs, strong adherence between the seal and the different cell components is required for the long-term stability of an SOFC. This paper reports on glasses with high CoO concentrations. CoO is commonly known as an adherence promoting oxide for improving the adhesion between metals (mostly steels) and vitreous enamels [24]. In the following, an attempt is introduced in which the effect of CoO in enamels is transferred to a crystallizing sealing glass of an SOFC. The CoO as a component of the glass composition is in a later stage incorporated into a crystal phase.

This paper reports on glasses based on the system $\text{BaO}/\text{ZnO}/\text{CoO}/\text{SiO}_2$. The densification, the crystallization behavior, the microstructure and the thermal expansion of the crystallized samples are described. Furthermore, the glass is used for joining a high temperature alloy suitable as interconnect for SOFCs.

2. Materials and methods

The glasses were prepared from reagent grade raw materials. Mixtures of SiO_2 , BaCO_3 (VK Labor- und Feinchemikalien), ZnO (Carl Roth GmbH+ Co. KG, Karlsruhe), Co_3O_4 (pure, VEB Laborchemie Apolda), ZrO_2 (E. Merck, Darmstadt), $\text{La}_2\text{O}_3 \cdot \text{H}_2\text{O}$ (VEB Laborchemie Apolda) and H_3BO_3 (E. Merck, Darmstadt) were used. They were melted in batches of 50g in a platinum crucible at a temperature of 1580°C , kept for 1h . Then the glasses were cast on a copper block and transferred to a furnace preheated to 700°C , which was subsequently switched off in order to allow the sample to cool. The batch compositions and the compositions determined by EDX-analysis (standardless measurement) of the studied glasses are given in Table 1.

Table 1. Glass compositions in Mol-%.

Sample	ZnO	CoO	BaO	SiO ₂	B ₂ O ₃	ZrO ₂	La ₂ O ₃
A (batch)	8.5	8.5	26	54	1	1	1
A (EDX)	9.1±1.7	9.8±1.0	24.3±1.1	55.4±1.8	– ^a	0.5±0.2	1.1±0.3
B (batch)	–	17	26	54	1	1	1
B (EDX)	–	17.15±1.3	23.03±1.2	58.4±2.5	– ^a	0.6±0.2	0.8±0.2

a

B₂O₃ could be detected but was not quantified.

The glasses were annealed at temperatures of 920 or 950 °C, kept for 1 h. The powdered (grain size <63 μm) and annealed samples were studied by X-ray powder diffraction (XRD), using a SIEMENS *D5000* diffractometer with Cu K_α radiation. The patterns were recorded in the 2θ-range from 10 to 60° with a step width of 0.02°. The XRD-patterns were analyzed using the software *DIFFRAC.EVA* from BRUKER.

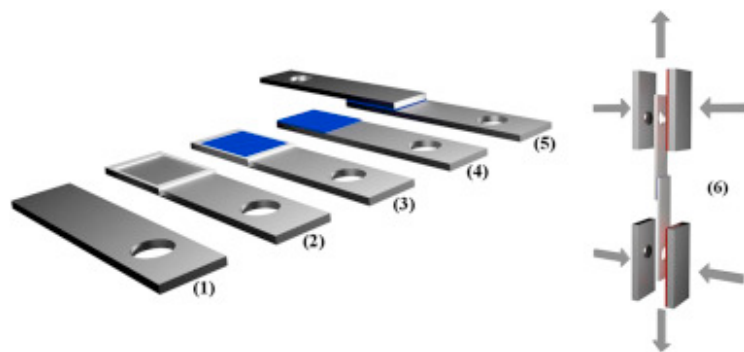
The dilatometric measurements (heating rate 5 Kmin^{−1}) were performed using a dilatometer NETZSCH *Dil 402 PC*. For the determination of the glass densities a MICROMERITICS *AccuPyc 1330 pycnometer* was used. The thermal properties were measured with a home-made DTA apparatus. For the determination of the sintering behavior a side-view hot-stage microscope was used. Using this device, images of the projection of a cylindrical shaped powder compact are made during heating the sample from room-temperature to a temperature above that attributed to the maximum densification.

To study the performance as a sealing material, the glass was powdered and sieved to grain sizes <71 μm. Then it was given to a cavity drilled in a plate (15×15×2.5 mm³) of a heat resistant alloy and slightly densified using a spatula. The alloy selected for the joining experiments was Nicrofer[®] 6025 HT – alloy 602 CA from ThyssenKrupp VDM. The metal with the powdered glass was heated up to temperatures between 865 and 945 °C using a heating rate of 5 Kmin^{−1}. The temperature was kept for 40 min. The sealing procedure was carried out in a silica tube with an inner diameter of 5.5 cm flushed with argon. The as prepared samples were cut and the cut planes were subsequently ground and polished.

The microstructures of the sealed samples especially near the interface metal/crystallized glass were studied using scanning electron microscopy (SEM Jeol *JSM 7001F*).

For the determination of the long-term behavior all the samples described above were prepared twice and were heated using a heating rate of 5Kmin^{-1} up to $(840\pm 10)^\circ\text{C}$, kept for 300h in air. The chosen temperature is a typical SOFC operating condition. The samples annealed for 300h are called A300 and B300.

In order to measure the strength of the composite, samples schematically shown in [Fig. 1](#), were prepared. The sealing temperature was 865°C kept for 40min. The mechanical testing carried out is schematically displayed in [Fig. 1\(6\)](#). For this purpose, a mechanical testing machine ZWICK 1445 was used.



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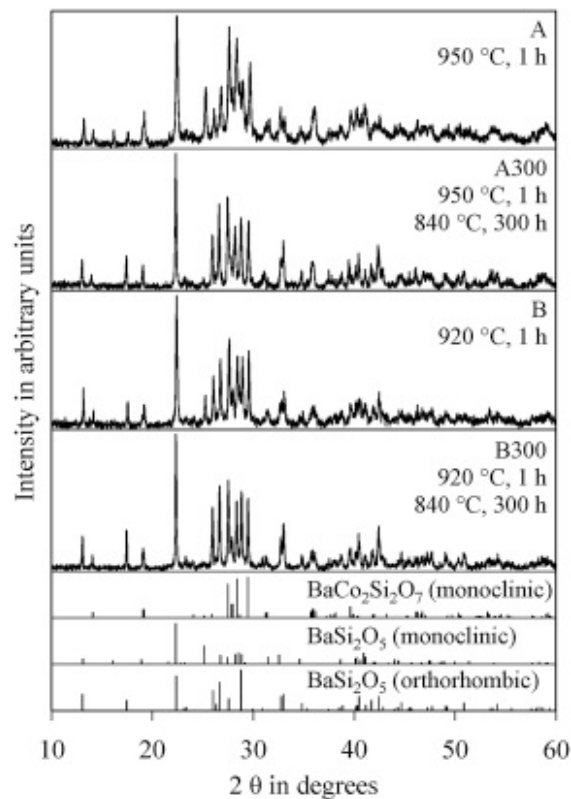
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Fig. 1. Preparation of samples and mechanical testing: (1) Metal plate consisting of Nicrofer[®] with a size of $(16\times 75)\text{ mm}^2$. (2) Metal plate with a mask consisting of aluminum on it. (3) Mask filled with glass powder ($<71\mu\text{m}$). (4) Removal of the mask. The glass layer has a height of 1 mm and a length of 20 mm. (5) Overlapping metal plate on the glass layer (6) Mechanical testing of the composites. The cylinders on the left jaws serve as a fixation of the samples. The red coating on the right jaws consists of rubber which protects the samples against excessive distortion during clamping and mechanical testing. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

3. Results and discussion

The glasses had black color and according to the recorded XRD patterns were amorphous. From thin sections of the glass UV-vis-NIR spectra were recorded and showed solely lines attributed to Co^{2+} , a broad absorption line attributed to Co^{3+} centered at around 380nm was not observed. Hence, despite of the used raw material, cobalt occurred entirely as Co^{2+} . The densities of the glasses A (zinc and cobalt) and B (only cobalt) were 4.01 and 3.98 gcm^{-3} , respectively. The glass transition temperature, T_g , determined by dilatometry, was 679°C for glass A and 660°C for glass B. The dilatometric softening temperatures were 725 and 695°C for glasses A and B, respectively. The measured DTA profiles showed an exothermic peak with an onset at 901 (glass A) and 862°C (glass B).

[Fig.2](#) shows XRD-patterns of samples A and B annealed at 950 and 920°C, respectively, as well as the samples A300 and B300 where an additional heating step at 840°C was performed. It should be noted that according to the DTA-profile, the exothermic peak occurs in sample B at a lower temperature than in sample A. Therefore, for better comparison, sample B and B300 were annealed at a 30K lower temperature (920°C).



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Fig.2. XRD-patterns of samples A, A300, B and B300 after the respective heat treatment. The three patterns in the lower part of the diagram show the most intense theoretical peaks of the respective crystal phases.

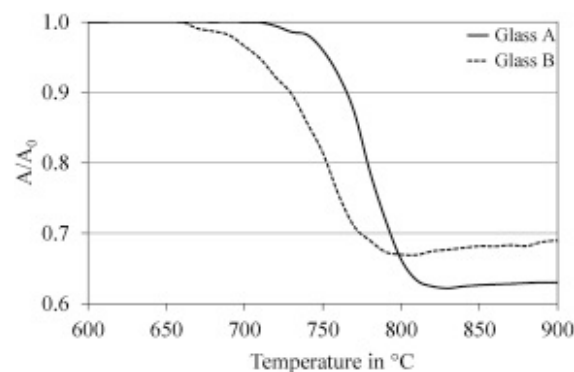
In the XRD-patterns of the samples A and B numerous distinct lines are observed which show a high degree of crystallinity.

In sample A, the crystalline phases $\text{BaCo}_2\text{Si}_2\text{O}_7$ (monoclinic, JCPDS no. 01-082-0184), BaSi_2O_5 (monoclinic, JCPDS no. 01-083-1446) as well as BaSi_2O_5 (orthorhombic, JCPDS no. 00-026-0176) were detected. The monoclinic BaSi_2O_5 is a high temperature phase, and according to the phase diagram is stable at temperatures above 1350 °C [25].

It should be noted that $\text{BaCo}_2\text{Si}_2\text{O}_7$ and $\text{BaZn}_2\text{Si}_2\text{O}_7$ form solid solutions $\text{Ba}(\text{Zn}_x\text{Co}_{2-x})\text{Si}_2\text{O}_7$ in the whole concentration range ($0 \leq x \leq 2$) [26]. Since the ionic radii of Co^{2+} and Zn^{2+} are nearly identical, also the lattice constants of the solid solutions do not depend much on the composition and hence it is scarcely possible to conclude from the XRD-patterns on the composition of the solid solution. Since sample B does not contain zinc, the attribution is clear, however, concerning sample A, it is highly probable that sample A contains a solid solution $\text{Ba}(\text{Zn}_x\text{Co}_{2-x})\text{Si}_2\text{O}_7$.

The XRD-patterns of the samples A300 and B300 do not show the peaks of the monoclinic high temperature modification of BaSi_2O_5 . This is most clearly indicated by the disappearance of the peaks at $2\theta = 16.1^\circ$, 21.6° , 25.2° and 32.6° . Hence, the long heat treatment at 840°C leads to the transformation of high temperature BaSi_2O_5 to low temperature BaSi_2O_5 .

Fig.3 shows the results from the hot stage microscopy recorded during densification of uniaxial pressed powders of the samples A and B. A_0 and A are the areas of the projection of the cylindrical powder compacts before the beginning of the sintering and during annealing, respectively. Densification starts at a temperature of 710°C for sample A and 660°C for sample B. At temperatures of 830 and 800°C , the densification is completed and the samples exhibit residual porosities of less than 5%, investigated by SEM micrographs.



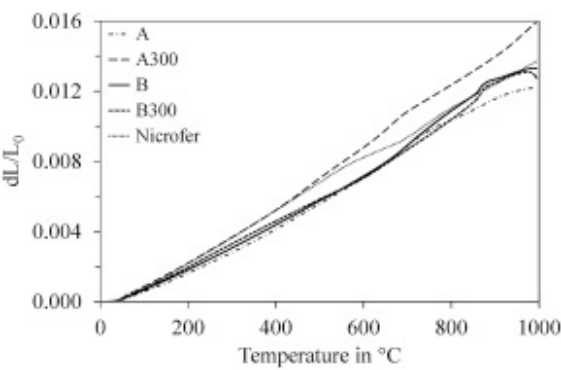
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Fig.3. Hot-stage microscopy of the glasses A and B.

Fig.4 shows dilatometric curves of samples crystallized at 950°C (A and A300) and at 920°C (B and B300) before and after a heat treatment of $(840 \pm 10)^\circ\text{C}$, kept for 300h. For comparison also the dilatometric curve of Nicrofer[®] is shown. In Table2, the CTEs for

different temperature intervals are shown for samples A, B, A300, B300 and the alloy. Within the temperature range from 100 to 800°C the CTEs of the crystallized glasses A, B and the alloy are all in the range from 14.0 to 14.6×10⁻⁶K⁻¹. The crystallizing seals fit well to the Nicrofer[®] alloy with respect to the thermal expansion.



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Fig.4. Thermal expansion of samples A and B, crystallized at 950 and 920°C, kept for 1 h. The samples A300 and B300 were prepared in the same way as the samples A and B, but they were subsequently heated up to 840°C, kept for 300h. The thermal expansion of the Nicrofer[®] alloy is also shown.

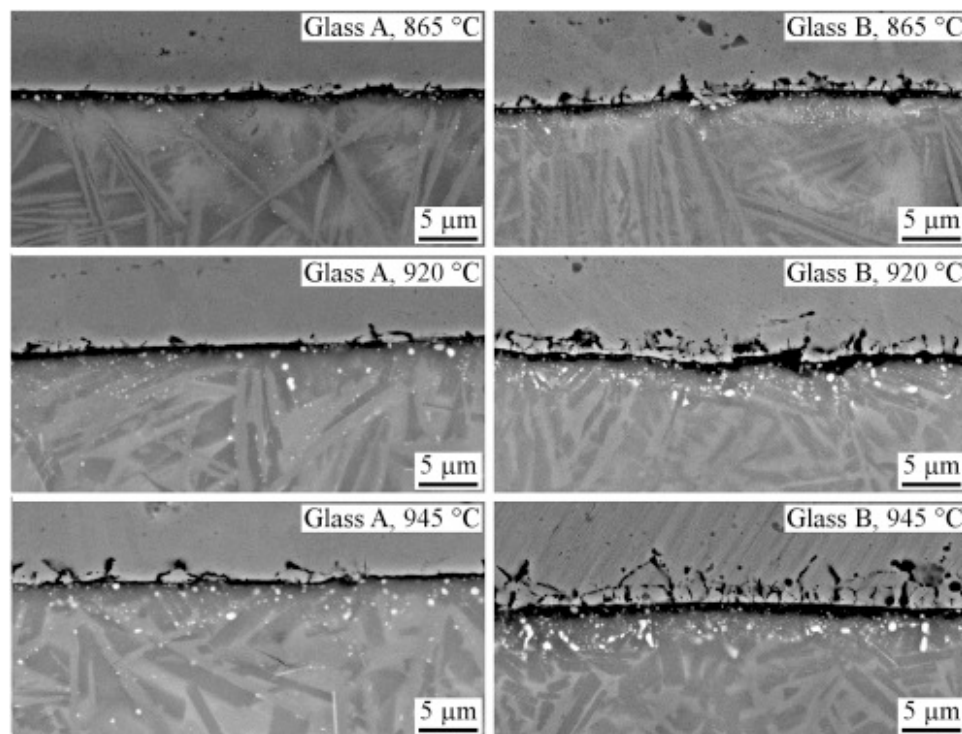
Table2. CTEs of the studied crystallized glasses and Nicrofer[®].

Temperature range in °C	CTE in 10 ⁻⁶ K ⁻¹				
	Nicrofer [®]	Sample A	Sample A300	Sample B	Sample B300
100–300	14.5	11.1	14.0	11.9	12.7
100–600	14.9	12.9	15.9	12.9	12.6
100–800	14.6	14.0	16.4	14.6	13.8
100–900	14.6	13.8	16.4	15.0	14.7

Temperature range in °C	CTE in 10^{-6}K^{-1}				
	Nicrofer [®]	Sample A	Sample A300	Sample B	Sample B300
100–1000	14.5	13.0	16.7	14.1	13.2

After 300h of heat treatment the dilatometric behavior of sample A has changed significantly (see sample A300 in [Fig.4](#)). Generally the trend is the same, but the dilatation is somewhat higher, resulting in CTEs being $(2.4\text{--}3.7)\cdot 10^{-6}\text{K}^{-1}$ higher than before the heat treatment. Most probably this is due to a more complete crystallization as well as to the above mentioned phase transition of high temperature BaSi_2O_5 to low temperature BaSi_2O_5 after 300h. Between room temperature and 500°C the thermal expansion behavior of sample A300 fits very well with the behavior of Nicrofer[®]. The thermal treatment at 840°C has only a minor effect on the dilatation of the sample B. The maximum difference of the CTEs of samples B and B300 is $0.9\cdot 10^{-6}\text{K}^{-1}$. This small difference in the CTE-values might be caused by different crystal orientations in the samples as well as the as mentioned phase transition [\[26\]](#). Furthermore, the steep increase in length of the samples B and B300 at around 850°C corresponds to a phase transition known from $\text{BaCo}_2\text{Si}_2\text{O}_7$ [\[26\]](#). This phase transition results in relatively small changes in the sample volume in comparison with similar crystal phases such as $\text{BaZn}_2\text{Si}_2\text{O}_7$ [\[19\]](#). However, this phase transition might induce stresses into the sealing material.

For sealing experiments, temperatures in the range from 865 to 945°C were chosen. The samples were subsequently cut, ground and polished and especially near the interface with the metal characterized using SEM. In [Fig.5](#), micrographs of samples prepared from the glasses A and B and the Nicrofer[®] alloy at temperatures of 865 , 920 and 945°C are shown. The upper part of each micrograph shows the metal, while the lower part shows the crystals precipitated from the glass and supposedly some residual glassy phase. In the lower part of all samples, near the metal numerous small spots of bright appearance are observed. The number of observed spots is higher in samples prepared from glass B than in those prepared from glass A. Increasing sealing temperatures also led to an increasing number of spots. The formation of the bright spots runs parallel to the formation of interconnected structures of dark appearance in the metal.



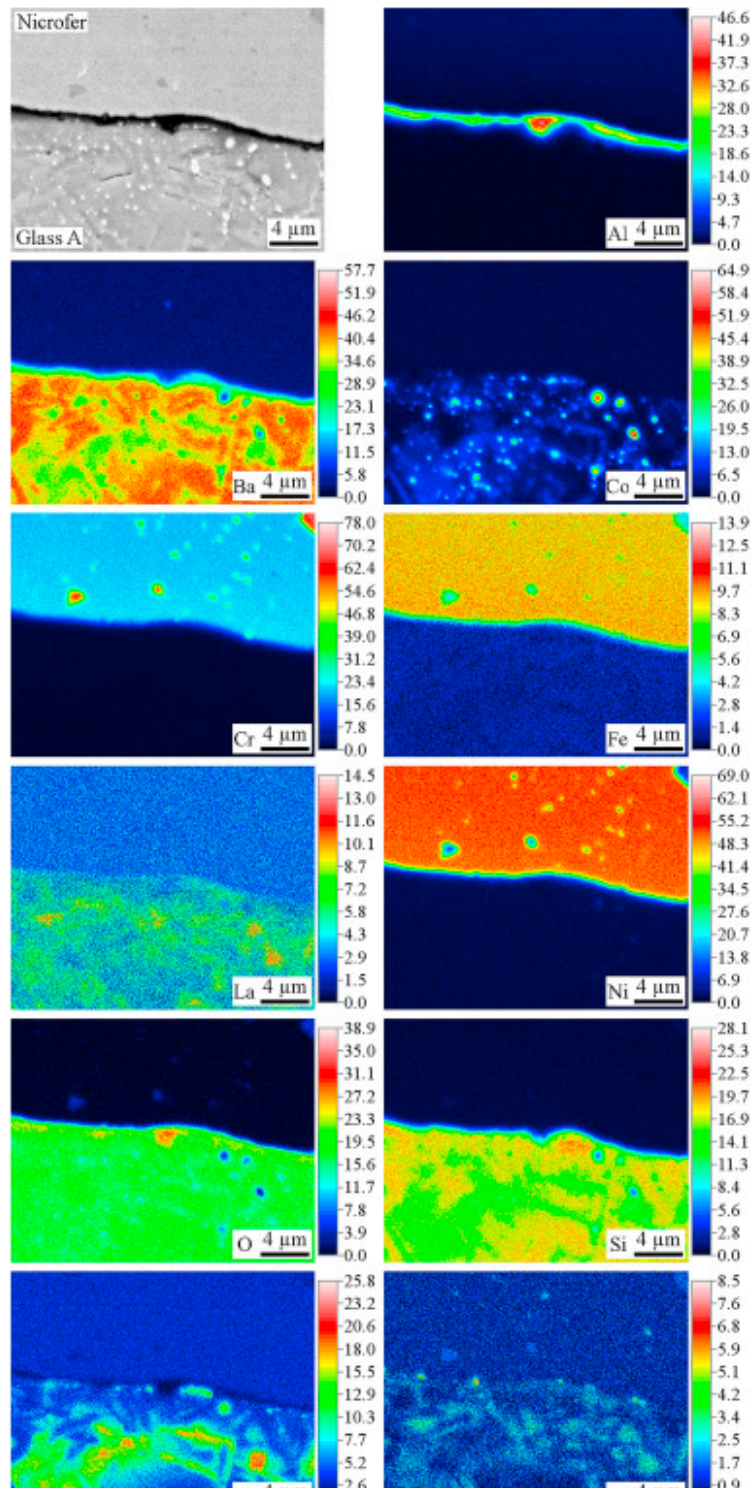
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Fig.5. SEM-micrographs of samples A (left) and B (right) sealed at 865, 920 and 945 °C (bottom: crystallized glass, top: metal).

Fig.6 shows EDX patterns of a sample prepared from Nicrofer[®] and glass A at 945 °C. The respective patterns are attributed to Al, Ba, Co, Cr, Fe, La, Ni, O, Si, Zr and Zn (the concentrations, see scale on the right, are given in wt%). The measured concentrations have a relative error of around 5%. The minimum uncertainties are around 0.1 wt%. The boundary between the metal and the crystallized glass appears dark and is strongly enriched in aluminum. Besides, notable quantities of oxygen occur in this phase. Hence it is mainly alumina. The bright spots already observed in the micrographs shown in Fig.5 are mainly composed of metallic cobalt; all other elements, including oxygen do not occur in notable concentrations in these spots. Furthermore, in the SEM, the particles appear even a little bit brighter than the Nicrofer[®] alloy and are hence definitely not an oxidic compound. Concerning the crystallized glassy phase, notable differences in the chemical composition occur: one phase is enriched in cobalt and zinc. Besides, notable quantities of silicon and barium as well as oxygen occur. The simultaneous occurrence of Zn and Co in

this sample is a strong hint of the formation of a solid solution [27]. This phase is most probably the solid solution $\text{Ba}(\text{Zn}_x\text{Co}_{2-x})\text{Si}_2\text{O}_7$ which gave rise to XRD-patterns similar to $\text{BaCo}_2\text{Si}_2\text{O}_7$. Another crystalline phase contains mainly barium, silicon and oxygen. This should be the BaSi_2O_5 phase observed in the XRD-patterns. Lanthanum and zirconium are depleted in those regions, the barium concentrations are highest.



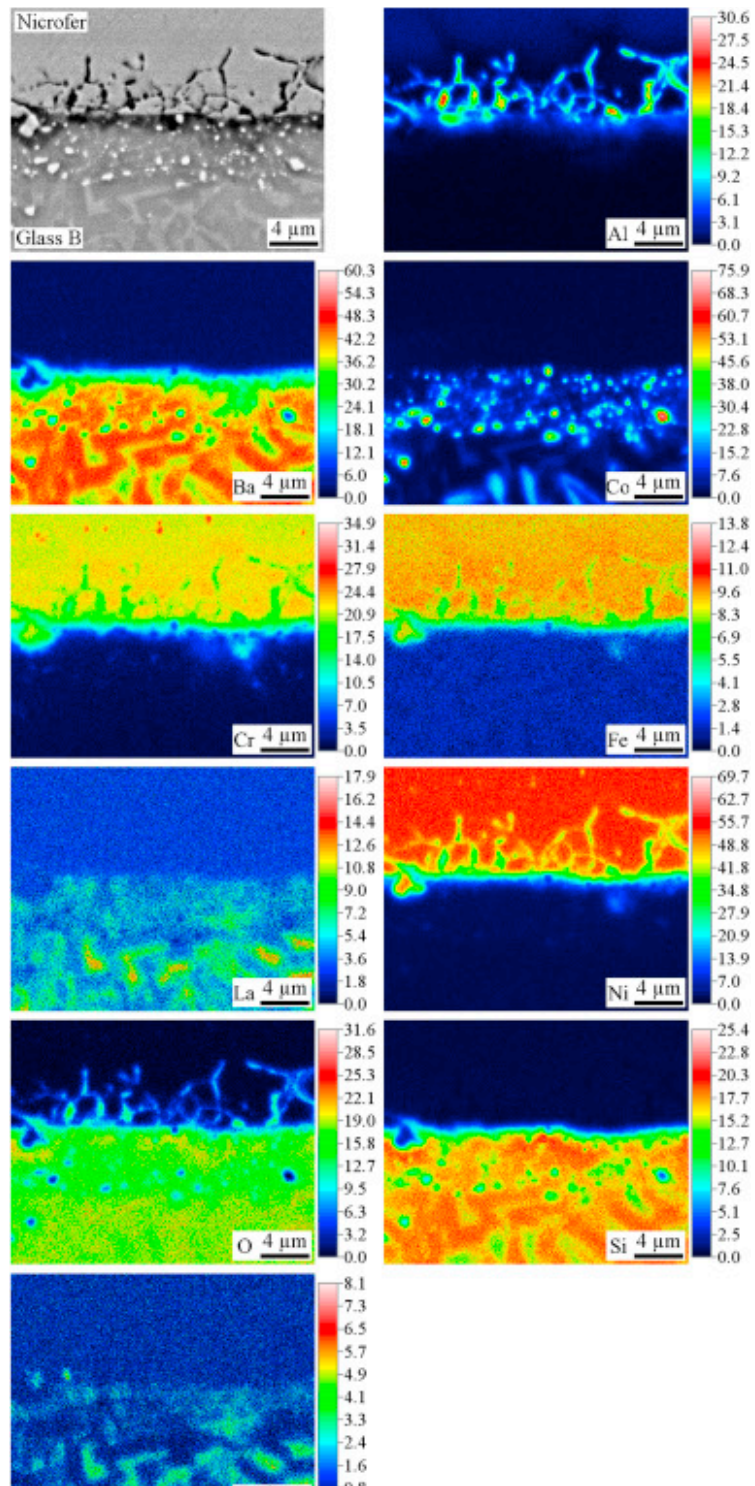


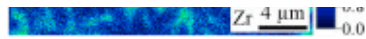
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Fig.6. SEM-micrograph and EDX mappings of sample A, sealed to the metal at 945 °C Al, Ba, Co, Cr, Fe, La, Ni, O, Si, Zr and Zn (the concentrations are given in wt%).

In Fig. 7, EDX-patterns of the sample prepared from Nicrofer[®] and glass B are shown. The detected elements are the same as in Fig. 6, (besides Zn, which does not occur in this sample). The bright spots are more numerous than in Fig. 6 and by analogy are composed of metallic cobalt. Within the limits of the method applied other elements are not detected within these bright spots. In comparison to Fig. 7, the interfacial layer does not occur in the same way. However, a strongly interconnected structure of dark appearance occurs in the metal near the boundary to the crystallized glass. This structure is strongly enriched in aluminum as well as in oxygen. In analogy to the boundary layer observed in Fig. 6, the interconnected structure seems to be alumina, however, due to the small size of this structure ($<0.5\mu\text{m}$), it cannot be excluded that also other elements occur in this phase. Furthermore, the chemical composition of the crystallized glass is not homogeneous within the whole volume. There are crystals which are strongly enriched in barium and silicon, while other parts of the microstructure are depleted in barium, but enriched in lanthanum, zirconia and cobalt. Near the interface to the metal, only minor concentrations of cobalt are incorporated into the oxidic crystals, because most of the cobalt occurs as metallic particles. In both samples, i. e. those prepared from sample A and from sample B, nickel (from the Nicrofer[®] alloy) scarcely occurs in the oxidic phase.





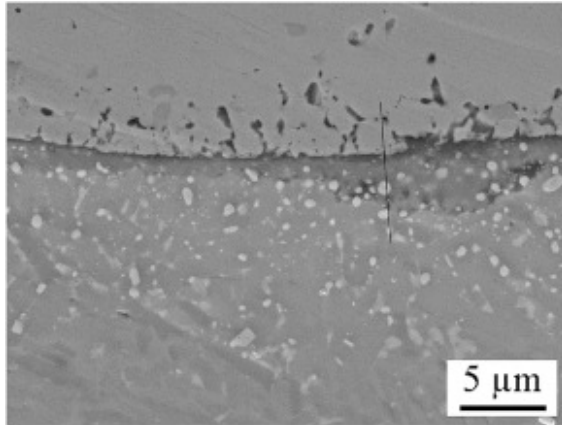
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Fig.7. SEM-micrograph and EDX mappings of sample B, sealed to the metal at 945°C Al, Ba, Co, Cr, Fe, La, Ni, O, Si and Zr (the concentrations are given in wt%).

The EDX-patterns in [Figs.6](#) and [7](#) show the interface between the glass-ceramic and the metal alloy. This interfacial layer causes strong adherence, which was measured, using the experimental setup, displayed in [Fig.1](#). On both metal plates, adherent glass is observed and the crack runs in between the crystallized glass and does not follow the interface metal/crystallized glass. That proves a strong adherence. The force necessary to fracture the sample was 780N, which related to the area of the seal is attributed to a strength of 2.7MPa.

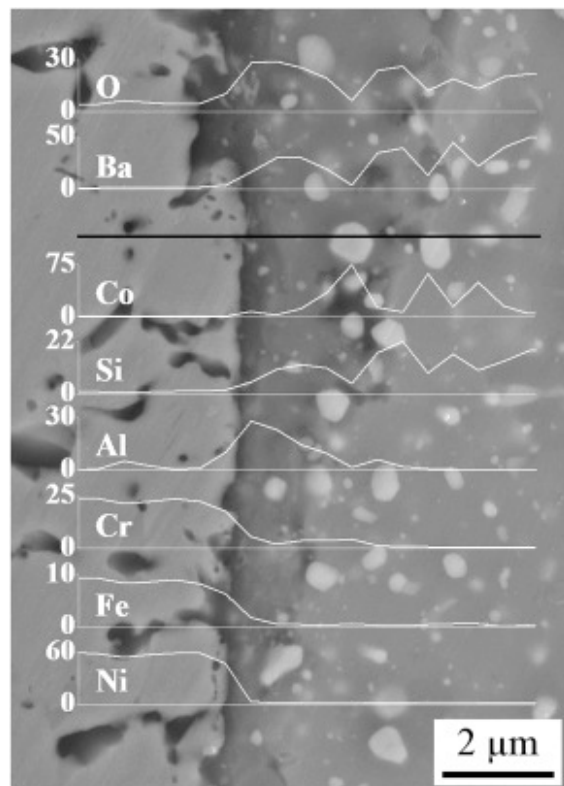
After a heat treatment of 300h at 840°C in air all the sealed samples were undamaged. [Figs.8](#) and [9](#) show SEM-micrographs and an EDX-analysis of one of the samples. This sample was prepared from glass B and the alloy. The sealing temperature was 945°C. So this sample is comparable with the sample from [Fig.7](#). The uncertainties of the EDX-line scan are the same as described for the EDX-mappings. The interface shown in [Figs.8](#) and [9](#) does not show any significant difference compared to the interface shown in [Fig.7](#). There is a slight diffusion of chromiumoxide and alumina into the glass. The dimensions of the cobalt particles and the diffusion zones are marginally affected by the thermal treatment at 840°C. Some cobalt particles occur in a larger distance from the interface in comparison with [Fig.7](#). This might be caused by the diffusion of small amounts of FeO into the glass, which is able to reduce the CoO to metallic cobalt.



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Fig.8. SEM-micrograph of the interface of glass B, sealed to the metal at 945°C and afterwards heat treated at 840°C for 300h. The black line marks the position of the line scan shown in [Fig.9](#).



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Fig.9. EDX-line analyses of the interface of glass B and Nicrofer[®]. The samples were sealed at 945 °C for 40min and afterwards heat treated at 840 °C for 300h. The line scan was performed along the black line. The y-axes show the concentration in wt%.

Both studied glasses can easily be densified by thermal treatment. The crystallization process obviously does not disturb the densification by viscous flow. During subsequent crystallization, BaSi_2O_5 as well as solid solutions $\text{Ba}(\text{Zn}_x\text{Co}_{2-x})\text{Si}_2\text{O}_7$ (in the case of sample A) or $\text{BaCo}_2\text{Si}_2\text{O}_7$ are formed. The thermal expansion coefficients are in the range suitable for sealing of nickel alloys such as Nicrofer[®]. The sealing process is very different from sealing processes reported up to now in the literature. Obviously a chemical reaction between the metal and the glass takes place. The aluminum of the metal (Nicrofer[®] contains approximately 2wt% Al) is oxidized while the cobalt oxide present in the glassy phase is reduced to the metal. In principle, this process is known from the enameling of metals such as steel [28]. By contrast to enameling, however, the glass is crystallized during the sealing

process which is a prerequisite for the high CTE necessary for the aimed applications. In the case of sample A which besides CoO also contains ZnO, an interfacial layer with some interconnected structures, is formed between the metal and the crystallized glass. In the case of sample B, at the same temperature, considerably more of such interconnected structures are formed in the metal. The interfacial layer as well as the interconnected structures are probably composed of alumina. Obviously, the adherence of the crystallized glass to the metal is strong as seen by the crack running through the crystallized glass during the fracture experiment.

4. Conclusions

Glasses based on the systems BaO/CoO/SiO₂ and BaO/ZnO/CoO/SiO₂ were studied with respect to their sintering and crystallization behavior. They showed good densification and during crystallization phases such as BaSi₂O₅ and Ba(Zn_xCo_{2-x})Si₂O₇ (in the case of sample A) or BaCo₂Si₂O₇ are formed. The crystallized samples exhibit CTEs (100–800°C) of $14.0\text{--}14.6 \times 10^{-6} \text{K}^{-1}$. Sealing experiments showed strong adherence of the crystallized glass to the Nicrofer[®] alloy. In SEM-micrographs it is seen that aluminum from the alloy is oxidized and forms an alumina layer or other structures inside the Nicrofer[®] alloy, while cobalt from the glass is reduced to the metal. This leads to the formation of approximately spherical cobalt particles occurring near the Nicrofer[®] alloy. The number of cobalt particles increases with increasing sealing temperature. During fracture of sealed Nicrofer[®] plates, the crack runs through the glass and not through the interface metal/crystallized glass. The thermal expansion coefficient of the crystallized seal can be adjusted by the cobalt concentration.

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